

- 3** A particle P is projected with speed 25 ms^{-1} at an angle θ above the horizontal from a point O on a horizontal plane and moves freely under gravity. After 2 s the speed of P is 15 ms^{-1} .

(a) Find the value of $\sin \theta$.

[5]

[illegible]

(b) Find the range of the flight.

[3]

[illegible]